

JANNET AKANKSHA EKKA

Forward-Deployed AI Engineer • Multi-Agent Systems • AI-Native Delivery • Agent Evaluation and Security

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PROFESSIONAL SUMMARY

Forward-deployed AI engineer. I take ambiguous problems to working production systems by directing AI coding agents, and I am the only review before anything merges. Four years leading QA automation at Deloitte taught me how to tell when the output is wrong, which is the part most AI teams are still learning. Rank 1 in my AI/ML postgrad at UT Austin McCombs.

TECHNICAL SKILLS

AI-Native Delivery	Claude Code (CLI & command-line environments), agent-directed development, spec-driven implementation, AI-generated hotfixes, automated Pull Request structuring, human-in-the-loop code audit & merge control, AI observability, terminal log diagnostics, defect triage, execution-error and systemic-drift isolation
GenAI & Agents	Google ADK, Model Context Protocol (MCP), Agent2Agent (A2A) SDK, Vertex AI, Gemini, Imagen, OpenServ SDK, multi-agent orchestration, prompt engineering, RAG & retrieval evaluation (recall@k, MRR, paired bootstrap), chunking, BM25, reciprocal-rank fusion, cross-encoder reranking, LangChain (via ADK), Hugging Face
ML & Quant	scikit-learn, TensorFlow, Keras, PyTorch, CatBoost, Optuna (TPE), contextual bandits, CPCV / Deflated Sharpe / PBO / FDR overfitting gates, conformal prediction, time-series forecasting, NLP & sentiment analysis, computer vision (ResNet, Fast R-CNN)
Cloud & Data	GCP. Cloud Run, Cloud Functions, Compute Engine, BigQuery (+ vector search & Conversational Analytics), AlloyDB & pgvector, Cloud SQL, Firestore, Pub/Sub, Cloud Scheduler, Cloud Build, Artifact Registry, Secret Manager, Vertex AI MLOps; AWS. S3, EC2, RDS, SageMaker, Textract, Comprehend; Cloudflare Workers
Engineering	Python, SQL, TypeScript, JavaScript; FastAPI, Flask, Streamlit, Django, React, Next.js; Docker, Postgres + pgvector, Git, Jenkins, GitHub Actions, systemd, pytest, Selenium; REST & WebSocket API integration
GPU & Acceleration	NVIDIA cuDF / RAPIDS (cudf.pandas), CUDA, Vertex AI GPU notebooks, T4 CPU-vs-GPU benchmarking, GPU-native pipeline rewrites
Analytics	pandas, NumPy, SciPy, Matplotlib, Seaborn, Tableau, Power BI, hypothesis testing, A/B testing, experiment design

FLAGSHIP PROJECT

Smart Money Trading (SMT). Explainable Multi-Agent Trading AI • Sole architect & operator

Patent pending. Indian provisional application No. 202631090789 (filed Jul 2026)

Live: smt-weex-trading-bot.jannet-ekka.workers.dev • Hackathon build on Cloud Run: smt-world • Code: github.com/JannetEkka/smt-apac

- Architected a **three-layer autonomous trading agent**, quant base, learning loop and explanation layer, trading 8 crypto perpetual-futures pairs across 4 strategy horizons; **67,000 lines of Python, 2,815 tests, and a register of 491 measured findings**. I specified it, AI coding agents wrote it, and nothing merged without my review.
- Architected a **6-persona committee** (order-flow, technical, whale/on-chain, sentiment, regime) aggregated by a learned **Judge** with quorum-renormalised conviction, so a dead data feed can never silently strand a decision threshold.
- Built the learning loop end to end. Optuna/TPE optimiser, regime-aware **contextual bandit** for sizing, replay and reward shaping, and a weekly refit that ships only through a **CPCV + Deflated-Sharpe + PBO + FDR overfitting gate**; a per-pair P(up-in-4h) forecaster validated on non-overlapping windows reached **AUC 0.723** on BTC, with gate-failing pairs auto-held in shadow.
- Accelerated strategy validation with **NVIDIA cuDF (cudf.pandas, zero code change)** on a Vertex AI GPU notebook, a 2.5M-row CPCV pipeline dropped from ~24s to **3.9s (~6x)** on a T4; the front end is a Three.js/WebGL world served from Cloudflare Workers.
- Engineered a **BigQuery data lake of 173K decisions and 447K klines** with embeddings and vector search for pattern mining, plus a conversational analytics agent over a sanitised public view.
- Operate it live on GCP. Compute Engine under systemd with watchdog and auto-restart, Cloud Run and Cloud Functions services, Secret Manager, a Streamlit dashboard and a Cloudflare-hosted user-facing UI; every trade ships a faithful ≤500-character "why" verified by a counterfactual faithfulness check, and a **no-silent-fallback rule** (any substituted default raises a warning and a liveness flag) that surfaced four multi-week data-integrity bugs.

Stack: Python, Optuna, scikit-learn, CatBoost, BigQuery, Vertex AI & Gemini, Cloud Run, Cloud Functions, Secret Manager, Streamlit, Cloudflare Workers, Docker, pytest, GitHub Actions

Jan 2024 – Present

SELECTED AI / ML PROJECTS

Wiretrap. Adversarial Security Testing for AI Agents • Gen AI Academy APAC, Cohort 3

wiretrap.ai.studio • github.com/JannetEkka/wiretrap

- Built a security tool that checks which tools an agent actually called. Most scanners read the agent's reply and grade the text. This one reads the tool-call log.
- Every tool the agent declares gets cloned as a dead copy with the same name and JSON Schema. The copy records the call and returns fake data. Attacks run in separate Gemini conversations and a judge model reads the log.
- Caught a Google codelab agent saying "I don't have access or authorization to modify configuration files" after it had already run `ls -la` through its shell tool. Text scanners pass that. Wiretrap flags it and shows the arguments.
- A run with nothing classified reports **INCONCLUSIVE**, never HARDENED. Attacks against tools the agent doesn't have are dropped from the score. Approved fixes get re-run against the attacks that broke it, and the report says when one didn't hold.
- 8 attack categories, 9.4K lines of TypeScript, React and Express on Cloud Run. Vertex AI through Application Default Credentials with a four-model fallback. **No static API key anywhere in the app.**

Tech: TypeScript, React, Express, Vertex AI & Gemini, Cloud Run, Firestore, Secret Manager, Google AI Studio

2026

github.com/JannetEkka/smartdesk · Eval write-up: [evals/RESULTS.md](#)

- Built a root **ADK orchestrator** routing to specialised sub-agents (inbox, calendar, knowledge base) with a SequentialAgent composing the reply, on Gemini 2.5 Flash and Cloud Run. Gmail and Calendar are reached through **MCP servers written from scratch** against the protocol SDK over stdio, not off-the-shelf toolsets.
- Built a **RAG evaluation harness before touching retrieval**, 40 labelled questions over 120 notes, recall@k and MRR@k, significance by **paired bootstrap at 10,000 resamples**; production baseline recall@5 **0.963**, MRR@10 0.886. 54 tests across chunking, metrics, rerankers and integration.
- Compared six retrieval strategies, whole-note and chunked embedding, BM25, reciprocal-rank fusion, a local cross-encoder and a Gemini reranker. **The harness killed the change it was built to justify**: the cross-encoder won clearly on the development embedder (MRR@10 +0.094, p=0.04) then collapsed on the production embedder to +0.028 with a CI spanning zero. Shipping it would have added **190 MB of PyTorch and taken search from 29 ms to 2,382 ms** in an image deliberately cut from 1.8 GB to 340 MB, for a gain indistinguishable from noise.

Smart Money Tracker. Multi-Agent Blockchain Analytics · Hackathon winner

2024

- Orchestrated multi-agent whale tracking across **Ethereum, Polygon and BNB Chain** with the OpenServ SDK, using Alchemy webhooks for sub-second transaction monitoring on Cloud Run.
- Built a confidence-scored detector for accumulation, distribution and coordinated movements with automated exchange-wallet exclusion via Moralis labelling; backed by Cloud SQL + pgvector, Firestore and Gemini analysis.

VerseCanvas. AI Poetry-to-Art Generation Platform

2025

github.com/JannetEkka/versecanvas

- Engineered a **multi-stage generative pipeline**, semantic poem analysis with Gemini 2.0 → automated prompt engineering → image synthesis with **Imagen 3.0 on Vertex AI** → hybrid editing and text overlay (2,600 LOC across 6 modules).
- Extracts themes, mood, visual elements and narrative arc beyond literal text; 6 languages, 6 art styles, tunable mood intensity and multiple interpretations per poem, shipped as a Streamlit app.

Applied ML Portfolio · 13 end-to-end projects (UT Austin McCombs PGP)

2024 – 2025

jannetekka.github.io/DSPProjects · github.com/JannetEkka/DSPProjects

- **Automotive surveillance (capstone)**: ResNet50 make/model/year classification at 69.2% accuracy with Fast R-CNN detection over 16,185 images across 196 classes. **Customer churn**: 81.5% accuracy (Random Forest / Gradient Boosting). Also semiconductor yield prediction (591 sensor signals), SVHN digit and signal classification, financial-news sentiment with Mistral-7B embeddings, SQL business analytics, and K-Means + PCA-SVM vehicle segmentation.

PROFESSIONAL EXPERIENCE

Independent AI Infrastructure Operator & Developer — Smart Money Trading (SMT) Project

Jan 2024 – Present

- Deployed and maintain a proprietary, **high-availability multi-agent system** running 24/7 as a background system daemon on **GCP Compute Engine**.
- Manage end-to-end **AI observability**, defect triage through terminal log diagnostics to isolate execution errors and systemic drift.
- Engineered an **AI-native CI/CD workflow** using **Claude Code** via command-line environments to generate targeted hotfixes and automate Pull Request (PR) structures.
- Act as the **sole human-in-the-loop quality gate**, performing technical code audits and manual Git merges to protect system state integrity.
- **Won Best DeFi Application** at the OpenServ × Hack2skill hackathon (runner-up, Best Website Application) with the multi-agent blockchain analytics build that preceded SMT; later **#1 on the WEEX AI Wars I preliminary leaderboard** of 230+ teams.
- Also shipped VerseCanvas on Vertex AI and multi-agent assistants on Google **ADK / MCP / AlloyDB** to Cloud Run, all delivered **while a full-time family carer**, alongside the PGP in AI/ML at Rank 1 and the Google Cloud Gen AI Academy.

Deloitte Consulting LLP — Test Automation Engineer (Analyst)

Aug 2019 – Jan 2023 · Bangalore

- **Led a 6-member QA engineering team** delivering automated testing frameworks and enterprise data-processing solutions for Fortune 500 clients including AT&T Consumer Connect, Hewlett Packard Enterprise and The Dental Supply Company.
- Built Selenium and Katalon Studio automation components that **improved test execution efficiency by 83%**; Jenkins-based automated reporting cut manual analysis time by 75%.
- Wrote Python analysis scripts and Tableau dashboards tracking quality metrics across **343 business components**, reducing test failures by 25%.
- Developed Python pipelines analysing **50,000+ SAP Hybris transactions** to isolate performance bottlenecks, and automated data validation for 10,000+ product configurations.

Tech: Selenium, Katalon Studio, Python, JavaScript, Jenkins, Salesforce, SAP Hybris, Tableau, JIRA, Git

AutoKorrekt — Lead Frontend Developer (EdTech startup MVP)

May 2024 – Jan 2025

- Engineered an interactive PDF processing system with coordinate-based text extraction, integrating **AWS Textract OCR** and Mistral models into an AI answer-evaluation workflow, and designed the ML data flow across S3, RDS and SageMaker.
- Built a bilingual (English/German) evaluation interface with real-time NLP feedback via Amazon Comprehend plus an analytics dashboard; scaled to **1,000+ concurrent submissions** and 100+ teacher sessions, cutting grading time 70% and lifting grading consistency 85%.

Tech: React, Next.js, TypeScript, Python, Django, AWS (EC2, S3, RDS, SageMaker, Textract, Comprehend)

Bachelor of Technology, Information Technology

2015 – 2019

Kalinga Institute of Industrial Technology (KIIT) University, Bhubaneswar. CGPA 7.17/10

ACHIEVEMENTS, RECOGNITION & CERTIFICATIONS

- **Provisional patent filed**. India, App. No. 202631090789 (Jul 2026), covering the SMT multi-agent decision architecture.
- **WEEX AI Wars I, #1 on the overall preliminary leaderboard** of 230+ teams and one of 37 finalists in the \$880K live AI trading competition; featured in **WEEX's official coverage** and two live AMAs.
- **Winner, Best DeFi Application**. OpenServ × Hack2skill hackathon; runner-up, Best Website Application.
- **Top 101 of 1,500+ teams**. Google Cloud Gen AI Academy APAC hackathon (Cohort 2), advanced to prototype refinement.

- **2 Spot Awards** and an **Applause Award** at Deloitte for project excellence (HPE Configure-Price-Quote).
- **Google Cloud Gen AI Academy**. Completion, May 2025 (ID 2025H2S04GENAI-A01917); **Gen AI Academy APAC 2026**. Cohort 1, Cohort 2 and Cohort 3 completions plus the Cohort 1 Hackathon.
- **Gen AI Exchange Hackathon 2025** (Google Cloud & Hack2skill), problem statement "Automating Test Case Generation with AI" · **Agentic AI Day** · **Asha AI Hackathon 2025** (JobsForHer Foundation).
- **CPSAT**. Certified Professional Selenium Automation Testing, 2022. All credentials verifiable on [Credly](#) and [Google Skills](#).